



qbio
quantitative
biology

QBIO MASTER PROGRAM

quantitative biology in practice

$$\frac{du}{dt} = \frac{\alpha_1}{1 + v^\beta} - u$$

Imaging Biological Systems

Philosophy and organization

Antoine Le Gall
Christine Doucet

antoine.le-gall@cnrs.fr
christine.doucet@cnrs.fr

What is in the black box ?



=> Build your setup



Advanced Microscopies

Introductory module

- laser safety / good practice
- basics practicals
- 3 hours seminar on optical microscopy, smFRET, N&B.

State of the art microscopy on biological samples - 4 groups

- experimental design
- 12 hours sample preparation + imaging + analysis
- **written + oral restitution (in groups)**

November

October

December

InLab1

- microscopy data analysis
- optical rays simulation

Build your setup - 4 groups

- design + troubleshooting
- 12 hours practicals
- **written + oral restitution (in groups)**



!! This is THE calendar to consider !!

Note that for the Advanced Microscopies practicals, there might be some adjustments of the schedule due to specific requirements for sample preparation.
You will be advised by your supervisors.

InLabs + Basics optics



Christine
DOUCET



Antoine
LE GALL



Caroline
CLERTE

Advanced microscopies



Rémi DORE



Christine
DOUCET



Jean-Bernard
FICHE



Caroline
CLERTE



Robert QUAST



Cédric
GODEFROY

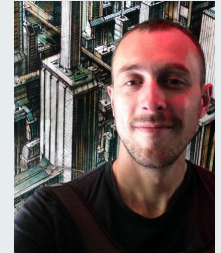
Build your setup



Simon
LETURQ



Malo MARMOL



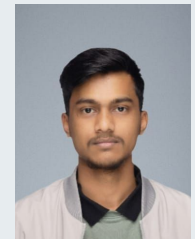
Jean-Bernard
FICHE



Emmanuel
MARGEAT



Antoine
LE GALL



Ajit
ROY

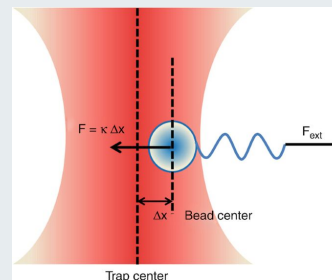
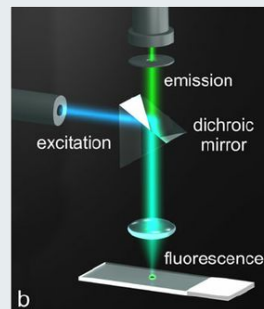
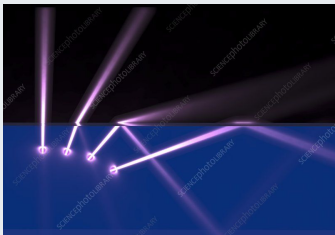


PART 2: BUILD YOUR SETUP



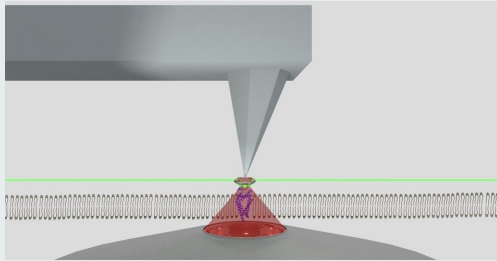
Choose 1 out of 4

1. Build an epifluorescence / TIRF microscope
2. Build a confocal microscope
3. Build an Optical Tweezer
4. Assemble a commercial microscope + metrology

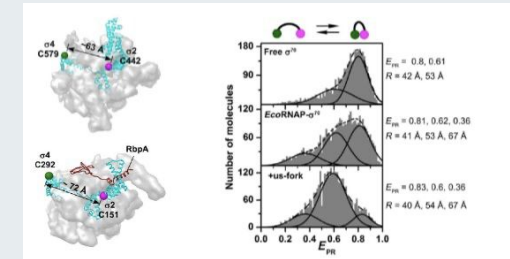


PART 3: ADVANCED MICROSCOPY

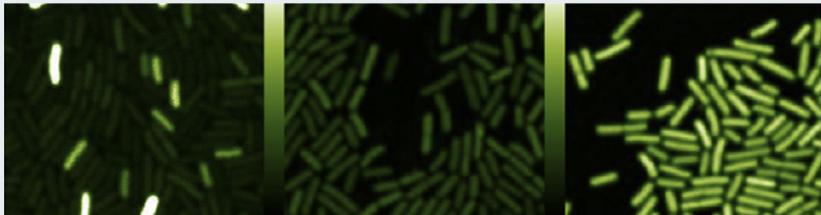
Choose 1 out of 4



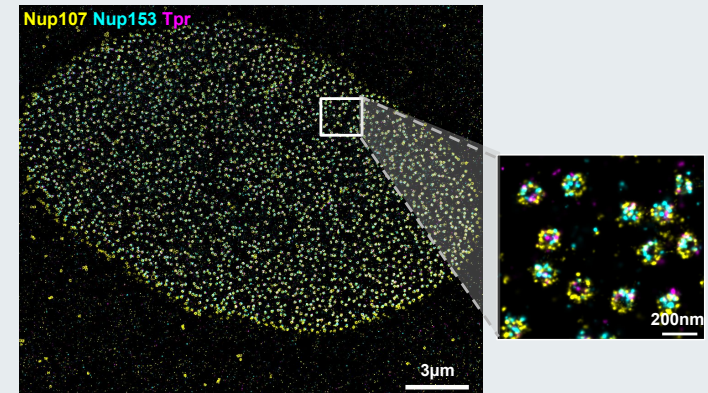
1- Image model lipid membranes by correlative AFM - Fluo



2- Structural dynamics of metabotropic Glutamate receptor by smFRET



3- Characterize promoter strength in *E.coli* by N&B



4- Nuclear Pore Complexes imaging in human cells using super resolution microscopy

To Do

- ❑ Constitute student pairs for the optics basics practicals (wednesday 7/10/26)

TP basics Optics

- ❑ Choose 1 out of 4 “build your setup” project + 1 out of 4 “advanced microscopies”

=> try, as much as possible, to vary partners and microscopy type.